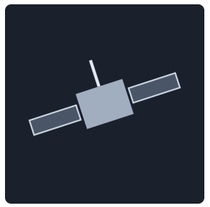
Locked recipe (DEC-018 / EXP-109 "C on A"): augmentation-trained Tesseract trees on frozen features + in-plane orbit PoE voting

frozen · trained (DINOv3 branch, both variants) · MoGe-2 branch: PoE variant only · training-free fusion · orbit inference (no parameters)

Training (synthetic/train only): arm A augmentation



GT square crop, label q

3 views per image (1 drawn per step)

- clean crop, label q
- 2 augmented copies (fixed seeds 1, 2):
  - photometric: brightness, contrast, gamma, noise, blur, glare blobs, random erasing, JPEG
  - in-plane rotation θ ~ U(−180°, 180°) about crop centre
  - $q' = \mathrm{Exp}(-\theta \cdot d) \otimes q$
  - d: ray through crop centre (PnP-checked)

DINOv3 ViT-L/16
tokens 8×8 × 1024
❄ frozen · 6.0 ms

MoGe-2 ViT-L
normals 16×16 × 4 (+ mask)
❄ frozen · 20.5 ms

Tesseract tree Transformer A (DINOv3)
LN+Linear → 256, 2-D pos. enc., 1 encoder layer; path decoder (2 layers)
[ROOT, $c_0 \ldots c_4$]: root → 4 charts, child → 8; depth 5 = 131,072 cells
loss: per-level cross-entropy with geodesic soft targets

Tesseract tree Transformer B (MoGe-2)
same architecture, separate weights; same augmented views
trained only for the PoE variant

Recipe facts

- trainable params: 2.39 M (tree A) + 2.13 M (tree B)
- budget: 100 epochs, batch 256, AdamW 1e-3, bf16
- training time: ≈ 13 min (A) / ≈ 6 min (B) per seed
- augmented features pre-extracted once (frozen backbones)
- selection: synthetic val + synthetic val-aug (DEC-006)
- lightbox / sunlamp never used for training or selection

Not in the recipe (tested, rejected): 16×16 tokens (EXP-109 B), weighted PoE (EXP-110), discrete path flow (EXP-108), distillation (EXP-044), Kurz corner tree (EXP-105 K).

Inference: in-plane orbit PoE voting (C)

test crop → 4 views

θ_k ∈ {0°, 90°, 180°, 270°}
rotated about the crop centre

per view k (same trained trees)

DINOv3 → tree A: p_A^(k)(chart, child | prefix)
MoGe-2 → tree B: p_B^(k) (PoE variant)

$$\log p^{(k)} = \log_\mathrm{softmax}(\log p_A^{(k)} + \log p_B^{(k)})$$

DINOv3-only variant: p^(k) = p_A^(k)
beam search → top-8 leaves

map back to the original frame

$c = \mathrm{Exp}(+\theta_k \cdot d) \otimes q_{view}$

4 views × 8 leaves = 32 candidates

(the exact inverse of the training label transform)

a wrong 180° flip rarely wins in all 4 views;
the true pose does

orbit score (product across views)

$s(c) = \Sigma_k \log p^{(k)}(\mathrm{leaf}(\mathrm{Exp}(-\theta_k \cdot d) \otimes c))$

teacher-forced tree likelihood of c
as seen from each rotated view

$\hat{q} = \mathrm{argmax}_c\ s(c)$

q̂ (leaf centre) — results, 3 seeds, mean °

variant	synth	lightbox	sunlamp
PoE + orbit (best)	**3.87**	**12.30**	**12.28**
PoE greedy (no orbit)	4.86	17.35	20.16
DINOv3-only greedy*	4.84	22.5	22.0
DINOv3-only + orbit	EXP-112 (pending)		

prior frozen EXP-036 (PoE greedy): 5.54 / 26.91 / 36.66

* single-branch numbers from the training-run evaluation

Cost (batch 1, shared GPU, measured components; end-to-end per variant measured in EXP-112)

DINOv3-only	DINOv3 6.0 ms	tree ≈ 6 ms (one tree, est.)		greedy ≈ 12 ms · with orbit ≈ 4× backbone + 4 beams + scoring
PoE (DINOv3 + MoGe-2)	DINOv3 6.0 ms	MoGe-2 20.5 ms (+ consensus mask)	both trees 12.8 ms	greedy ≈ 39 ms · with orbit ≈ 4× (est. 120–150 ms)

Batched (64): DINOv3 410 img/s, MoGe-2 41 img/s, both trees 4,489 img/s: MoGe-2 dominates the PoE variant's cost; orbit voting multiplies the backbone cost by 4.

Overhead of MoGe-2 = (PoE − DINOv3-only) in ms and in degrees; overhead of orbit = (orbit − greedy). Both are reported per variant in EXP-112.

Frozen backbones: 303 M (DINOv3-L) + 331 M (MoGe-2) parameters, not trained; trainable: ≈ 2.4 M (DINOv3-only) / ≈ 4.5 M (PoE).

Rotation only, ground-truth crops; selection on synthetic validation (DEC-006); lightbox / sunlamp test-only. Severe flips (> 150°) corrected by PoE + orbit vs EXP-036: 62% (lightbox, of 401) and 80% (sunlamp, of 206); good predictions broken: 0.4%.

Sources: EXP-109 RUN-20261016-014754 (accuracy, flip transitions), EXP-107 timing (components), DEC-018 (recipe lock). EXP-112 adds the DINOv3-only + orbit accuracy and end-to-end timing for all four variants.